Custom Exam Paper (Mark Scheme)

Part	**Answer**	**Marks**	**Partial Marks / Guidance**
1(a)	$6\sqrt{2}$	2	M1 for $\sqrt{6^2+6^2}$
1(b)	$3\sqrt{2}$	1	B1 for $\frac{1}{2}\times 6\sqrt{2}$
1(c)	45°	2	M1 for $\tan(\angle VPM)=\frac{3\sqrt{2}}{3\sqrt{2}}$
2(a)	Show proof	3	M1 for $\frac{30}{x}-\frac{30}{x+5}=\frac{1}{2}$
			M1 for $30(x+5)-30x=\frac{1}{2}x(x+5)$
			A1 for establishing $x^2+5x-300=0$ with no errors
2(b)	2 hours	3	M1 for $(x+20)(x-15)=0$ or correct substitution into formula
			A1 for $x=15$ (rejecting $x=-20$)
			A1 for time $=\frac{30}{15}=2$
3(a)(i)	27	1	B1
3(a)(ii)	$\frac{1}{2}$ or 0.5	1	B1
3(b)	$4x^7$	2	B2 for correct answer
			M1 for $16x^{12}$ seen
4(a)	$x=20$	2	M1 for $4x+(x+30)+(x+30)=180$ or $6x+60=180$
4(b)	80°	1	A1 for 4×20
5(a)	$\frac{3}{5}$	2	M1 for converting to improper fraction $\left(\frac{25}{9}\right)^{-\frac{1}{2}}$ or finding $\sqrt{\frac{9}{25}}$
5(b)	4	2	M1 for writing all terms with base 3: $\frac{3^{-5}\times(3^2)^2}{(3^3)^{-1}}$ or $3^{-5+4-(-3)}$ or 3^2

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6	$x = \frac{5}{y-1}$	3	M1 for clearing denominator: $yx = x+5$
			M1 for isolating and factorising: $x(y-1) = 5$
7(a)	$\frac{240}{360} \times \pi \times 3^2 = 6\pi$	2	M1 for $\frac{240}{360} \times \pi \times 3^2$
7(b)(i)	30 GBP	2	M1 for 6×5
8(a)	$18\frac{4}{7}$	3	M1 for total distance $= 36$, M1 for total time $= \frac{12}{16} + \frac{24}{20} = \frac{3}{4} + \frac{6}{5} = \frac{39}{20}$ hours, A1 for $\frac{36 \times 20}{39} = \frac{240}{13}$ or $18\frac{6}{13}$
8(b)	4.6	2	M1 for mass $= 4000 \times 1.15 = 4600$ g
9(a)	56	1	Angles in same segment
9(b)	42	1	Angles in same segment
9(c)	98	2	M1 for $\angle AEB = 180 - 42 - 56 = 82$, then $180 - 82$
10(a)	$y = -2x + 10$	2	M1 for substituting $(3,4)$ into $y = -2x + c$ or $y - 4 = -2(x-3)$
10(b)	$(0,10)$	1	B1 strictly for coordinates format
11(a)	$(3,3)$	2	M1 for $\left(\frac{-3+9}{2}, \frac{8-2}{2}\right)$
11(b)	Correct proof shown	2	M1 for gradient of $AB = \frac{-2-8}{9-(-3)} = -\frac{5}{6}$
			A1 for perpendicular gradient $= \frac{6}{5}$ and substituting $(3,3)$ to show $6x - 5y = 3$
11(c)	$\left(\frac{1}{2}, 0\right)$	2	M1 for setting $y = 0$ in the equation
12(a)	1000	3	M1 for writing individual values to 1 sig fig as 40, 5, and 0.2, M1 for $\frac{40 \times 5}{0.2}$ or $\frac{200}{0.2}$, A1 for 1000.
13(a)	$10x - 7$	2	M1 for $6x - 15$ or $4x + 8$
13(b)	$6x(2x + 3y)$	2	M1 for $3x(4x+6y)$ or $6(2x^2+3xy)$ or $2x(6x+9y)$

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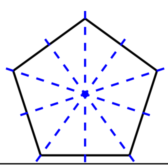
13(c)	$(x-4y)(x+4y)$	2	M1 for $(x-4y)(x-4y)$ or similar error
14(a)	$50000\times 0.8^2 = 32000$	2	M1 for 50000×0.8^2 or $50000\times\left(\frac{4}{5}\right)^2$ or 40000 seen
14(b)	25600 EUR	2	M1 for 32000×0.8 or 50000×0.8^3
15(a)	$\frac{1}{8}$	1	B1 for correct fraction simplified completely
15(b)	2.6	1	B1
16			
	$x>1$	1	Accept strict inequality only
	$y\geq 1$	1	
	$y<-x+6$	1	Accept $x+y<6$
17*	8 cm	3	M1 for area ratio $\frac{18}{32}=\frac{9}{16}$
17*			M1 for linear scale factor $\sqrt{\frac{9}{16}}=\frac{3}{4}$ or $\frac{4}{3}$
18(a)	$(1,-1)$	2	M1 for $\left(\frac{-2+4}{2},\frac{-5+3}{2}\right)$ or one coordinate correct
18(b)	Radius $=\frac{1}{2}\sqrt{(4-(-2))^2+(3-(-5))^2}=\frac{1}{2}\sqrt{6^2+8^2}=\frac{1}{2}(10)=5$	2	M1 for finding diameter $=10$ or using distance from centre to C or D
19(a)		2	B1 for 1 to 4 correct lines and no extras
19(b)	5	1	
20(a)(i)	$x<3$	2	M1 for $x+1<4$

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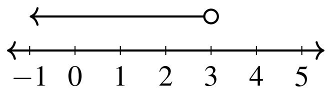
20(a)(ii)	$-2 \leq x < 3$	2	M1 for $-4 \leq 2x$ or $2x < 6$
20(b)	$-2, -1, 0, 1, 2$	2	B1 for one missing or one extra integer
20(c)		1	Open circle at 3 and arrow to left
21(a)	9 litres per minute	2	M1 for $\frac{150 \times 60}{1000}$ or equivalent
21(b)	Show that height increases by 3.6 mm	3	M1 for converting volume per hour: $9 \times 60 = 540$ litres or 0.54 m^3
			M1 for height $= \frac{\text{volume}}{\text{area}} = \frac{0.54}{2.5} = 0.216$ m
			A1 for converting 0.216 m to 216 mm and dividing by 60 to get 3.6 mm or equivalent complete proof
22(a)	$y = 7$	1	B1 for correct equation
22(b)(i)	$\left(-\frac{2}{3}, 0\right)$	2	M1 for setting up equation $y + 5 = -\frac{3}{4}(x - 6) \Rightarrow 3x + 4y = -2$
			A1 for setting $y = 0 \Rightarrow x = -\frac{2}{3}$
22(b)(ii)	$\left(0, -\frac{1}{2}\right)$	2	M1 for setting $x = 0$ in their equation
			A1 for $y = -\frac{1}{2}$
23(a)	67.5%, $\frac{17}{25}$, 0.682, $\left(\frac{1}{8}\right)^{-1} 2 \times 10^{-1}$		M1 for evaluating individual terms to equivalent forms: $0.68, 0.675, 0.682, 0.8$
23(b)	$=$ or $\geq$ or $\leq$	1	B1 for verifying both sides are structurally equivalent to 0.035
24(a)	150 EUR	3	M2 for $180 \div \frac{120}{100}$ or $180 \times \frac{100}{120}$ or M1 for identifying $120\% = 180$
25(a)	55	2	M1 for $\frac{1}{2} \times (8 + 14) \times 5$
25(b)	$\frac{1}{2} \times 11 \times h = 55$	1	M1 for correctly equating area expression to 55
	$h = 10$	1	A1 for clean algebraic isolation without errors

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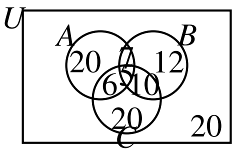
26(a)	-9	2	M1 for $2\times 9-9\times 3$ or $18-27$
26(b)	-105	2	M1 for $10\times 2-125$ or $20-125$
27(a)	300 g	2	M1 for $\frac{120}{8}\times 20$ or 15×20
27(b)	30	2	M1 for $\frac{450}{15}$ oe
28(a)	5, 10, 20	1	B1 for all three terms correct
28(b)	160	1	B1
29(a)	$\mathrm{n}(A\text{ only})=40-4x$ $\mathrm{n}(B\text{ only})=37-5x$ $\mathrm{n}(C\text{ only})=\mathrm{n}(\text{none})=40-4x$ Sum: $(40-4x)+(37-5x)+(40-4x)+(2x-3)+(x+1)+2x+x+(40-4x)=155-11x=100$	3	M1 for expressing individual single regions in terms of x M1 for full summation expression equal to 100 A1 for flawless verification reaching $155-11x=100$
29(b)	 Left: 20, Right: 12, Bottom: 20, Top middle: 7, Bottom-left middle: 6, Bottom-right middle: 10, Centre: 5, Outside: 20	3	B1 for solving $x=5$ and finding the four intersection regions B1 for correct single-sport regions (20, 12, 20) B1 for correct outer region (20)
30(a)	$6(x+1)-2x=x(x+1)\Rightarrow 4x+6=x^2+x\Rightarrow x^2-3x-6=0$	3	M1 for multiplying by $x(x+1)$, M1 for expanding brackets correctly
30(b)	$x=5, x=-2$	3	M1 for $x^2-3x-10=0$, M1 for $(x-5)(x+2)=0$
31(a)	1.5×10^3	2	M1 for 15×10^2 or 1500
32*	2	2	M1 for $\frac{5-1}{2-0}$ or equivalent
33(a)	$10\sqrt{3}$	2	M1 for $12\sqrt{3}$ or $2\sqrt{3}$

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33(b)	$7-3\sqrt{3}$	3	M1 for multiplying numerator and denominator by $2-\sqrt{3}$
			M1 for numerator $10-5\sqrt{3}+2\sqrt{3}-3$
34(a)	8	1	
34(b)	$16\sqrt{3}$	2	M1 for $\frac{1}{2}\times 8\times 8\times \sin(60^\circ)$
34(c)	Equal chords are equidistant from the centre	1	Accept unambiguous equivalents
35(a)	-11	2	M1 for $3-8=-5$ or $5\times(-5)$
35(b)	8.5	2	M1 for $1.2\div 0.3=4$
36(a)	13 20	1	B1 for subtracting 2 hours from departure time.
36(b)	18 20 on 13 April	2	M1 for adding 3 days to the date (13 April) and 5 hours to the time ($1320+5$ hours).
37(a)	420	2	M1 for prime factorisations $42=2\times 3\times 7$ and $60=2^2\times 3\times 5$, or for listing multiples
37(b)	19:37	2	M1 for converting 420 seconds to 7 minutes A1 for adding 7 minutes correctly to 19:30